

Host-listener dialogue (Topic: Uncanny valley effect)

Host: Listener:

大家好，欢迎收听本期播客。今天我们要聊一个非常有趣且富有启发性的心理学现象——恐怖谷效应。它不仅关系到机器人设计，还深刻影响我们与拟人化技术的互动体验。

Hello everyone, and welcome to this podcast. Today we're going to talk about a fascinating and thought-provoking psychological phenomenon—the uncanny valley effect. It not only relates to robot design but also profoundly influences our interaction experience with human-like technology.

嗯，听名字就挺吓人的，“恐怖谷”？这到底是什么呢？能不能先给我讲讲它的基本概念？

Hmm, the name sounds pretty scary. What exactly is the “uncanny valley”? Could you start by explaining its basic concept?

当然可以。恐怖谷效应是指，当机器人或者其他非人类物体的拟人化程度越来越高时，人类对它们的情感反应并不是线性增加的。最初，我们会觉得它们很可爱或者有亲和力，但当它们看起来几乎像人，却又有些怪异时，我们反而会感到厌恶甚至害怕，这个阶段就叫“恐怖谷”。之后，如果拟人化程度再进一步接近真人，我们的好感又会回升。

Of course. The uncanny valley effect refers to the phenomenon where, as a robot or other non-human entity becomes more human-like, people's emotional response doesn't increase linearly. At first, we may find it cute or appealing. But when it looks almost human yet slightly off, we feel disgusted or even frightened—this stage is called the “uncanny valley.” Later, as it becomes even closer to a real human, our positive feelings rise again.

哦，原来是个曲线变化啊。那为什么会出现这种突然的反感呢？有研究解释这个现象吗？

Oh, so it's a kind of curve. Why does this sudden aversion occur? Have there been any studies explaining this phenomenon?

有的。这个效应最早是由日本机器人专家森政弘提出的。背后的心理机制有几个假说，比如进化假说。它认为人类大脑进化出一种自我保护机制，当我们看到“几乎像人但又不完全对劲”的形象时，会把它们当成潜在的疾病或者基因异常，进而产生厌恶和回避的反应。

Yes. This effect was first proposed by the Japanese robotics expert Masahiro Mori. Several hypotheses have been proposed to explain its psychological mechanisms, such as the evolutionary hypothesis. This hypothesis suggests that the human brain has evolved a self-protection mechanism. When we see something that is “almost human but not quite right,” we may perceive it as a potential sign of disease or genetic abnormality, which triggers disgust and avoidance.

哇，原来是进化遗留下来的“警报系统”啊。那这个假说有实验支持吗？

Wow, so it is like an “alarm system” left over from evolution. Is there experimental support for this hypothesis?

有的。比如研究人员用猴子做实验，发现猴子更愿意看真实的猴脸或者不太逼真的人造猴脸，而对非常逼真的人造猴脸反而不太感兴趣，说明这种“恐怖谷”反应可能不仅是人类独有。

Yes. For example, researchers conducted experiments with monkeys and found that monkeys preferred looking at real monkey faces or less realistic artificial monkey faces, but showed less interest in highly realistic artificial monkey faces. This suggests that the “uncanny valley” response may not be unique to humans.

嗯，挺有意思的。除了进化假说，还有别的解释吗？

That is quite interesting. Besides the evolutionary hypothesis, are there any other explanations?

还有一个叫做“违背预期假说”。它认为，当我们看到一个外观非常像人的机器人，但它的动作却不符合我们对人的预期时，大脑会感到矛盾和不适，从而引发恐怖谷效应。研究用脑成像技术发现，观看仿真机器人动作时，大脑活动更活跃，说明我们在努力调和这种不匹配。

There is also the “expectation violation hypothesis.” It suggests that when we see a robot that looks very human-like but moves in a way that does not match our expectations of human behavior, the brain experiences conflict and discomfort, which can trigger the uncanny valley effect. Brain imaging studies have found that when people observe the movements of humanoid robots, brain activity becomes more intense, suggesting that we are trying to reconcile this mismatch.

所以说，恐怖谷效应其实是多种因素共同作用的结果，是进化和认知双重机制的体现？

So, the uncanny valley effect is actually the result of multiple factors working together, reflecting both evolutionary and cognitive mechanisms?

正是如此。视觉造型、触感、动作等多感官因素都会影响这种反应。

Exactly. Multi-sensory factors such as visual appearance, tactile qualities, and movement can all influence this reaction.

那在设计机器人或者数字虚拟人时，这个效应有什么启示吗？

What implications does this effect have for designing robots or digital avatars?

有很重要的启示。设计师不能盲目追求高度拟人化，因为一旦触碰到恐怖谷，用户的反感可能会急剧增加。比如在数字虚拟人中，如果外观非常真实，但声音明显是合成的，就会加剧不适感。相反，采用卡通风格或者拟人化的动物形象，既能引发共情，又能避免落入恐怖谷。

It has very important implications. Designers should not blindly pursue a high degree of human likeness, because once a design falls into the uncanny valley, users' negative reactions may increase sharply. For example, in digital avatars, if the appearance is highly realistic but the voice is clearly synthetic, the discomfort may become even stronger. In contrast, using a cartoon style or anthropomorphic animal characters can evoke empathy while avoiding the uncanny valley.

嗯，原来不一定越真实越好，那人形机器人呢？

So, more realism is not necessarily better. What about humanoid robots?

对于人形机器人，除了面部外观的优化，还要注重肢体语言和情感交互的整体构建。比如“具身机器人”就强调“手眼心脑”的协同工作：眼睛负责交流眼神和表情，“心”感知用户情绪，“脑”则支撑深入的知识沟通，这样能提升亲和力。

For humanoid robots, in addition to optimizing facial appearance, designers also need to focus on the overall construction of body language and emotional interaction. For example, “embodied robots” emphasize the coordination of the “hands, eyes, heart, and brain”: the eyes support eye contact and facial expressions, the “heart” perceives users' emotions, and the “brain” supports deeper knowledge-based communication. This can improve the robot's perceived friendliness and approachability.

听起来很复杂，但也很智能。恐怖谷效应还能应用在哪些领域吗？

That sounds complex, but also very intelligent. In what other areas can the uncanny valley effect be applied?

一个很有趣的应用是人脸伪造检测。通过收集那些引发不适感的图像，比如僵尸或者玩偶，训练模型识别伪造人脸时产生的整体违和感，提高检测准确率。

One interesting application is facial forgery detection. By collecting images that evoke discomfort, such as zombies or dolls, models can be trained to recognize the overall sense of incongruity produced by forged faces, thereby improving detection accuracy.

哇，这个技术应用真是跨界了。那设计师在不同场景下，如何应对恐怖谷效应呢？

Wow, that is really a cross-disciplinary application. How should designers respond to the uncanny valley effect in different scenarios?

这就需要考虑使用环境的差异，比如公共场合和私人空间中，用户的情感联结不同。设计师要定制相应的交互策略，确保人机互动更自然、更符合情境，最终实现人机共融。

This requires considering differences in usage contexts. For example, users may form different emotional connections in public spaces and private settings. Designers need to customize interaction strategies to ensure that human-machine interaction feels more natural and contextually appropriate, ultimately supporting human-machine integration.

嗯，听你这么一说，恐怖谷效应不仅是个心理学现象，更是设计和技术创新的重要指南。

After hearing your explanation, I feel that the uncanny valley effect is not only a psychological phenomenon, but also an important guide for design and technological innovation.

没错，它提醒我们在技术进步中，必须尊重人类的心理边界，找到“似人”与“非人”之间的平衡。

Exactly. It reminds us that in technological development, we must respect human psychological boundaries and find a balance between being human-like and non-human.

谢谢你的分享，今天学到了好多。以后看到那些拟人机器人，我会更注意观察它们是否落入了“恐怖谷”。

Thank you for sharing this. I learned a lot today. In the future, when I see anthropomorphic robots, I will pay more attention to whether they fall into the “uncanny valley.”

很高兴你喜欢这期内容。希望大家也能带着好奇心，继续探索科技与心理的交汇点。我们下期再见！

I am glad you enjoyed this episode. I hope everyone can continue exploring the intersection of technology and psychology with curiosity. See you next time.